

Echo-Memory: A Controlled Study of Memory in Action World Models

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Abstract

We present **Echo-Memory**, a controlled study of memory mechanisms in action-conditioned world models. These models generate multi-segment videos from a first frame, text prompt, and camera-action sequence, but their central failure is often memory rather than local image synthesis: after the camera leaves and returns, the scene or salient object may silently change. Existing memory designs are hard to compare because gains are entangled with backbone, training, retrieval, and evaluation differences. Echo-Memory fixes the action-to-video interface and varies only how history is stored and read by the generator. Under a shared video diffusion backbone, optimizer, camera-action representation, sampler, and evaluation pipeline, we compare raw context, compression-based memory, spatial summaries with different read-out paths, and state-space recurrence. This matched matrix separates four otherwise conflated axes: capacity, compression, read-out, and recurrence. We also evaluate memory through a three-branch protocol: replay quality, in-domain loop revisit, and open-domain return probes. The branches routinely disagree, showing that replay fidelity is not a sufficient proxy for remembering a world. Three findings follow. Raw context is a strong capacity baseline and improves open-domain return far more than it improves replay metrics. Compactness is not a free substitute for capacity: aggressive spatial and hybrid-compression memories lose the salient evidence needed for return. Finally, block-wise state-space recurrence is the strongest open-domain return mechanism in our matrix, showing that the structure of implicit memory matters as much as the decision to use it. These results provide a compact protocol for studying memory in action world models beyond isolated replay metrics.

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Project Page: <https://echo-team-joy-future-academy-jd.github.io/Echo-Memory/>

Code: <https://github.com/Echo-Team-Joy-Future-Academy-JD/Echo-Memory>

1 Introduction

A central frontier for video generation is the move from producing a single plausible clip to producing a controlled world rollout. An action world model receives a first frame, a text prompt, and a sequence of camera actions. It must generate the next chunk, then the next, and continue doing so while preserving geometry, object identity, and camera obedience across revisits. The visible failures are familiar: the camera returns to the starting pose but the scene has silently changed, the salient object is replaced by a plausible impostor, or a chunk boundary wipes away the accumulated context. At their root, these are not generic generation failures. They are memory failures, and they are the failure mode that decides whether a system has actually modelled a world rather than merely extrapolated a clip.

Recent work has proposed several ways to supply this missing memory. Context-based methods retrieve pre-

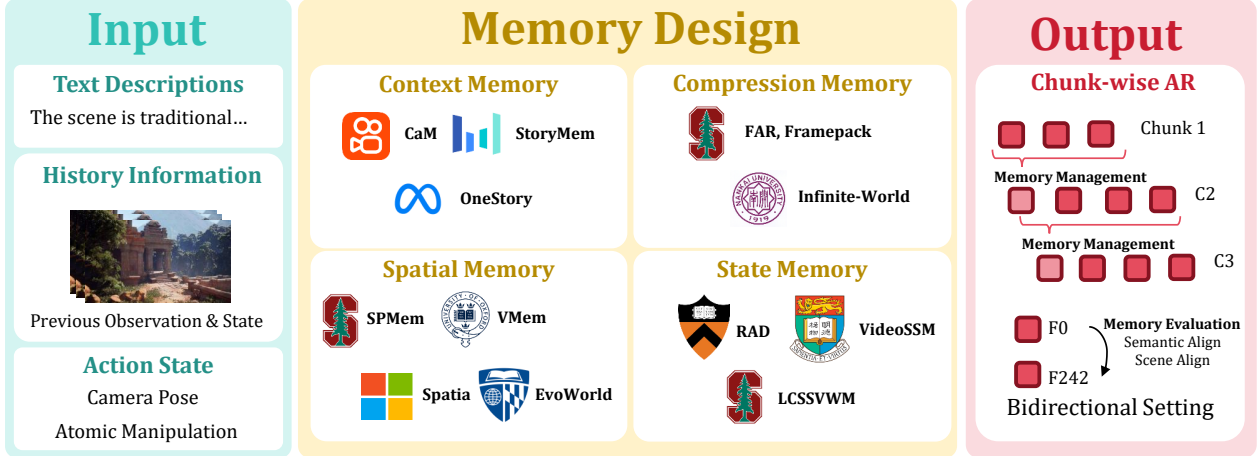


Figure 1 Abstract teaser and workflow of Echo-Memory. Given a text description, historical observations, and the camera/action state, an action world model must generate chunk-wise video while carrying memory across revisits. The figure positions Context, Compression, Spatial, and State-Space families as representative designs for preserving a revisitable world.

vious observations and append them as visual evidence for the current segment [24, 31, 57, 62]. Compression-based methods reweight or shorten the retrieved history to reduce cost [5, 28, 58, 71]. Spatial memories replace the full temporal stack with compact scene tokens [20, 25, 55, 57, 68], whereas state-space variants carry history implicitly through recurrence [9, 34, 44, 64, 66]. The same pressure appears in plug-and-play memory tokens, multi-shot keyframe memories, real-time interactive world models, persistent-world editors, and efficient long-form video diffusion systems [3, 7, 11, 39, 40, 45–49, 51, 56, 61, 65, 70]. These studies make clear that memory is now a central design axis for action-conditioned video generation. They also make comparison difficult: reported gains are often entangled with changes in backbone, training recipe, context budget, sampling procedure, and evaluation protocol.

This paper studies memory from a controlled, vision-centric perspective. Rather than introducing another memory module, Echo-Memory uses a fixed video diffusion-transformer backbone, data interface, camera-action encoding, sampler, and evaluation pipeline, and varies only the memory representation. This setting turns memory design into a common interface question: what information is stored, how is it compressed, how is it read by the generator, and whether it can survive a return motion after the camera leaves the visible support. The resulting comparison is intentionally narrower than the space of all world models, but it exposes differences that are otherwise easy to hide behind unrelated system changes.

Several methodological concerns motivate this design. Larger raw context may improve performance simply because it exposes more evidence, not because it provides a better memory mechanism; compression can appear efficient when token cost is hidden inside an operator whose retained evidence is never measured; and in-domain trajectories can overstate memory quality, because familiar textures allow plausible local reconstruction even when object identity is lost under distribution shift. We therefore evaluate every memory row through the same three branches: ground-truth replay, in-domain loop closure, and open-domain return. This also forces a separation between storage and read-out: a memory can store evidence without making it accessible to the generator, and one of our spatial read-out ablations shows that this distinction changes replay quality without improving revisit consistency.

Our study is organized around four memory families and three evaluation branches. The main empirical results are as follows:

- **Raw context is a strong capacity baseline.** Increasing context from $K=1$ to $K=20$ improves the replay image bundle and raises open-domain VLM return from 12.25 to 58.63.
- **Compact memory is not automatically semantic memory.** Spatial Memory is competitive on replay

PSNR but weak on open-domain return, while hybrid compression loses much of the signal preserved by simpler length compression.

- **Read-out matters as much as storage.** Storing tokens is insufficient unless the generator can use them at the moment of return; withheld and dedicated spatial read-out variants expose this difference.
- **Structured recurrence is the strongest current open-domain bias.** Block-wise State-Space memory reaches an open-domain VLM score of 69.00 despite lower replay PSNR, illustrating why replay alone is not an adequate memory score.

The rest of the paper develops this study. [Section 2](#) reviews memory mechanisms and evaluation practice; [section 3](#) defines the shared interface; [section 4](#) and [section 5](#) describe optimization and metrics; [section 6](#) reports the controlled comparisons; and [section 7](#) discusses limitations and future directions.

2 Action World Models: Preliminaries and Related Work

The key components of an action world model are the video diffusion backbone, the action representation and its injection site, the memory subsystem, the retrieval policy, the training schedule, and the evaluation and benchmarking protocol. Each component has its own intricacies, and the interactions among them are non-trivial. Our study investigates them under a single backbone and a single training budget, with the explicit goal of isolating the contribution of memory.

World models have been studied as latent imagination, predictive state dynamics, and joint-embedding prediction [21, 22, 29]. Recent video and embodied systems extend this view from compact latent states to generated visual rollouts conditioned on prompts, camera motion, or agent actions [17, 26, 33, 36, 63, 72]. This shift makes memory a practical modeling question: a rollout must not only synthesize the next plausible clip, but also preserve scene and object evidence after the camera leaves and later revisits a region.

Diffusion-Transformer Video Backbones. A modern video generator typically follows the latent-diffusion recipe: an autoencoder maps each frame to a low-dimensional latent, and a denoising network learns the conditional velocity field over a stack of latent frames. Recent work has converged on diffusion-transformer backbones [52, 53], which scale favorably and natively expose per-frame token positions. Compared to U-Net-based 3D diffusion models, DiT backbones are particularly well suited to action-controlled generation: any per-frame signal (camera RT, action labels) can be attached to the corresponding frame token and propagated through self-attention, and the choice of injection site directly determines how strongly the signal reaches the spatial attention. Throughout this paper we use an off-the-shelf video DiT [53] as the common substrate; all design-space choices we study are local edits to this substrate. The broader video-generation ecosystem has converged on increasingly powerful diffusion-transformer backbones, including Imagen Video [23], Sora [41, 43], Veo [14], Wan [52, 53], Seedance [19], Waver [67], joint audio-video DiT [35], and camera-controllable generative re-rendering [4, 37]. Native multimodal models such as Emu3.5 [13] explore unified token-based modeling across images, video, and language. Cosmos is better described as a world-foundation-model platform for Physical AI, with video foundation models for prediction and conditional world generation rather than a single native multimodal model [1, 2].

Action and Camera Conditioning. A second axis, orthogonal to the memory subsystem, is the representation of action signals. Camera-controllable video diffusion has explored a spectrum of representations: world-frame extrinsics in context-based action-world models [62], Plücker embeddings, dense optical flow, and discretized action tokens for game-style control as in game-style world models. Time-aware positional encoding [60], cube-map and panoramic parametrizations [32], and iterative video editing on the same scene [30] further illustrate alternative action and viewpoint representations. We adopt the most straightforward representation, 12-dimensional relative-RT camera poses, and treat the injection site as a first-class design variable: post-attention, pre-norm, pre-QKV, or gated pre-QKV ([section 3.2](#)). This mirrors a finding from the multimodal-LLM literature: a seemingly small plumbing decision (e.g., where exactly the connector reads from) often dominates large architectural ones.

Memory for Long-Horizon Video Generation. Long-horizon video generation has increasingly treated

memory as an interface between the current denoising window and observations that are no longer visible. One direct strategy is retrieval: previous frames or features are selected by camera overlap, entity identity, temporal proximity, or scene geometry, and then appended as context for the next chunk [24, 25, 31, 62, 68]. This line has been extended to streaming and interactive settings, where the system must decide what to keep in a fast cache, what to compress, and when to refresh evidence during rollout [10, 27, 48, 56]. Other work reduces the cost of carrying history by weighting tokens, packing temporal windows, pooling latent frames, or replacing a full frame stack with compact spatial summaries [5, 28, 57, 58]. Keyframe memories and multi-shot generation systems make a similar trade-off at the level of scenes or shots rather than individual tokens [3, 40, 65]. These methods differ in implementation, but they share the same underlying trade-off: the model must preserve enough visual evidence for a later return while keeping the current generation window affordable.

A complementary direction avoids explicit retrieval at inference time and carries history in a recurrent or learned state. State-space video world models propagate a causal hidden state through selected DiT blocks, whereas recurrent gates, test-time fast weights, flow-equivariant states, and plug-and-play memory tokens offer lighter or more modular variants [9, 34, 47, 64, 66]. Persistent-scene editors and spatial world memories similarly emphasize that what matters is not merely whether information is stored, but whether the generator can read it back when the viewpoint returns [11, 18, 45, 46, 50]. We use this literature to define a controlled comparison rather than a new memory module: all rows share the same backbone, camera interface, training budget, sampler, and evaluation protocol, while the memory representation and read-out path are varied.

Evaluation and Benchmarking. Static image and single-clip video metrics such as PSNR, SSIM, LPIPS, FID, and FVD remain the lingua franca of video generation, but on their own they do not capture long-horizon revisit consistency. Context-based action-world-model work has begun reporting specialized revisit metrics on rotation loops [24, 27, 48, 62], while the broader vision-language community has shifted toward model-as-judge protocols using strong VLMs (e.g., GPT-4V or Qwen3-VL-30B-A3B [54, 59]). Multi-shot storytelling benchmarks [3, 40, 65, 70], streaming long-video efficiency benchmarks [61, 69, 71], and out-of-sight dynamics evaluation [8, 16] further enrich this evaluation landscape. Spatial reasoning benchmarks and general VLM/LLM backbones also inform model-as-judge protocols for semantic verification [6, 12, 15, 38, 42, 59]. Our evaluation (section 5) combines both: long-horizon GT replay for PSNR/SSIM/LPIPS image quality, GT-backed in-domain 180° loop closure with revisit PSNR and VLM scoring, and open-domain 45° return probes measured by revisit PSNR proxies and VLM scores for SCENE and SPECIAL IDENTITY consistency.

How Echo-Memory differs. Our contribution is neither a new storage mechanism, nor a new retriever, nor a new conditioning trick. It consists of (i) a common interface into which all of the above can be plugged without modifying the training loop, sampler, or evaluation protocol; (ii) a matched ablation protocol that exercises the full design space under a single-stage training budget; and (iii) a three-branch evaluation stack that links trajectory replay, in-domain revisit, and open-domain revisit consistency. Rather than introducing yet another isolated model, the study is intended to make the interactions among storage, compression, read-out, recurrence, and evaluation visible within a single experimental matrix.

3 Method: The Echo-Memory Design Space

This section formalizes the memory design space studied by Echo-Memory. The point is not a new backbone, but a clean factorization of how an action world model can represent the past: Context tokens, Compression operators, Spatial summaries, or a State-Space state. All rows share the same training path and the same evaluation path; the row name changes only the memory/context profile.

3.1 Preliminaries

Backbone. Let $\mathbf{x}_{1:T} \in \mathbb{R}^{T \times H \times W \times C}$ be a target video segment of T frames at spatial resolution $H \times W$. A pre-trained video diffusion-transformer [52, 53] models the conditional velocity field $v_\theta(\mathbf{z}_t; t, \mathbf{c}_{\text{text}}, \mathbf{c}_{\text{ctx}}, \mathbf{c}_{\text{act}})$, where $\mathbf{z}_t$ is the noised VAE-encoded latent at flow-matching timestep t , $\mathbf{c}_{\text{text}}$ is a text embedding, $\mathbf{c}_{\text{ctx}}$ is a memory context, and $\mathbf{c}_{\text{act}} \in \mathbb{R}^{T \times 12}$ is a per-frame relative-RT camera-action sequence. The single-stage

training objective is the rectified-flow regression loss applied only on the target frames:

$$\mathcal{L}(\theta) = \mathbb{E}_{t, \mathbf{x}, \mathbf{c}} \|v_\theta(\mathbf{z}_t; t, \mathbf{c}_{\text{text}}, \mathbf{c}_{\text{ctx}}, \mathbf{c}_{\text{act}}) - v^*\|_2^2, \quad v^* = \mathbf{z}_1 - \mathbf{z}_0. \quad (1)$$

Per-frame VAE context. We use a per-frame VAE context representation, which produces one latent token group per video frame. This keeps the temporal dimension of $\mathbf{c}_{\text{ctx}}$ semantically aligned with the target tokens, and makes compression-based memory operations well defined as transformations on the temporal axis of a fixed-length latent stack.

Action representation. We use a 12-dimensional relative camera RT with 9 rotation entries and 3 translation entries, expressed in the reference frame of the first frame of the segment. Every action in this study is relative; absolute action encodings are out of scope.

Context-conditioned generation. For a current segment with start frame s and end frame e , the model receives the target segment together with a context set of K historical observations:

$$\mathcal{C}_{s:e} = \{(\mathbf{x}_{f_k}, \mathbf{p}_{f_k})\}_{k=0}^{K-1}, \quad f_0 = s, \quad f_{1..K-1} \subset \{0, \dots, s-1\}, \quad (2)$$

where $\mathbf{p}_{f_k}$ denotes the associated camera pose/action metadata. The first context element is the anchor observation for the current segment; the remaining $K-1$ elements come from a retriever $\mathcal{R}$ over historical frames. With probability $p_{\text{drop}} = 0.1$ we drop the retrieved frames and keep only the anchor, simulating a cold-start regime.

3.2 Memory Design Space

Echo-Memory treats memory as a pair of operations attached to an otherwise fixed video DiT. The retriever first forms the same evidence set for every row:

$$\mathcal{E}_{s:e} = \{(\mathbf{x}_s, \mathbf{p}_s)\} \cup \mathcal{R}_{\text{FOV}}((s+e)/2, \{(\mathbf{x}_j, \mathbf{p}_j)\}_{j < s}, K-1). \quad (3)$$

The memory profile π then decides only how this evidence is written and how the generator can read it:

$$\boxed{\mathbf{M}_i^\pi = W_\pi(\mathbf{M}_{i-1}^\pi, E(\mathcal{E}_i)), \quad \mathbf{U}_i^\pi = R_\pi(\mathbf{M}_i^\pi)} \quad (4)$$

$$\hat{v}_i = v_\theta(\mathbf{z}_t; t, \mathbf{c}_{\text{text}}, \mathbf{U}_i^\pi, \text{MLP}_\phi(\text{RT}_{1:T})). \quad (5)$$

This is the whole abstraction: all rows share the evidence $\mathcal{E}$, action embedding, sampler, and loss; they differ only in W_π and R_π . **Notation.** $E(\cdot)$ encodes selected frames into VAE tokens. $\mathbf{M}_i^\pi$ is the stored memory after segment i , and $\mathbf{U}_i^\pi$ is the object exposed to the DiT: context tokens, spatial tokens, or recurrent state. W_π is the write rule and R_π is the read rule. In Algorithm 1, α denotes token weights, Pool_r shortens time by ratio r , Grid_G writes a $G \times G$ spatial summary, ρ is the spatial read-out policy, and $\text{Rec}_\mathcal{B}$ updates the recurrent state in block set $\mathcal{B}$. The four memory families can therefore be read as simple choices of what survives between chunks: raw context keeps past frames nearly unchanged, compression keeps fewer or reweighted tokens, spatial memory keeps a scene-level token grid, and state-space memory keeps an implicit hidden state rather than explicit visual tokens.

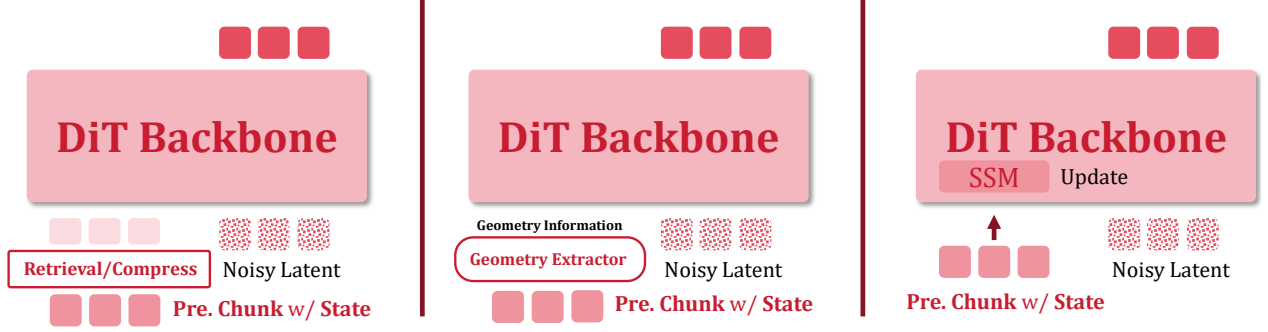


Figure 2 Overview of four representative approaches to memory in action world models. Under a shared video diffusion backbone and a shared camera-action interface, the approaches differ only in how historical information is stored and read. We compare four families throughout: Context, Compression, Spatial, and State-Space.

Algorithm 1: Memory profile for one segment

```

Input  $\mathcal{H}_{<i}, x_i, \text{RT}_{1:T}, \pi$ 
1  $\mathcal{E} \leftarrow \{x_i\} \cup \mathcal{R}_{\text{FOV}}(\mathcal{H}_{<i}, K-1)$ 
2 if  $\text{train} \wedge \text{rand}() < 0.1$  then  $\mathcal{E} \leftarrow \{x_i\}$ 
3  $\mathbf{Z} \leftarrow E(\mathcal{E})$ 
4 case  $\pi$  of
    ANCHOR:  $\mathbf{M} \leftarrow \mathbf{Z}_{\text{anchor}}; \mathbf{U} \leftarrow \mathbf{M}$ 
    RAWCTX:  $\mathbf{M} \leftarrow \mathbf{Z}_{1:K}; \mathbf{U} \leftarrow \mathbf{M}$ 
    COMPRESS:  $\mathbf{M} \leftarrow \text{Pool}_r(\alpha(\mathbf{Z}) \odot \mathbf{Z}); \mathbf{U} \leftarrow \mathbf{M}$ 
    SPATIAL:  $\mathbf{M} \leftarrow \text{Grid}_G(\mathbf{Z}); \mathbf{U} \leftarrow \rho(\mathbf{M})$ 
    STATESPACE:  $\mathbf{M} \leftarrow \text{Rec}_B(\mathbf{M}_{\text{prev}}, \mathbf{y}); \mathbf{U} \leftarrow \mathbf{M}$ 
5  $\mathbf{a} \leftarrow \text{MLP}_\phi(\text{RT}_{1:T})$ 
Return  $v_\theta(\mathbf{z}_t; t, \mathbf{c}_{\text{text}}, \mathbf{U}, \mathbf{a}), \mathbf{M}$ 

```

3.3 Concrete Variants

The representative rows instantiate the four families above under the same backbone and action interface: anchor-only I2V, raw Context at different lengths, Compression, Spatial Memory, and State-Space recurrence. Focused ablations then vary one mechanism at a time: Spatial read-out, Compression type, recurrence structure, or raw context length. The empirical tables introduce the exact rows where they are needed, so the method section can stay focused on the shared interface rather than on a separate registry.

Default configuration. Unless otherwise stated, every row uses relative-RT camera conditioning, a fixed flow-matching timestep shift, target-frame-only supervision, and the same inference-time memory profile as its training configuration.

3.4 A Unified Memory Interface

To keep the comparison fair, the training loop, sampler, and evaluation pipeline expose the same memory interface to every variant. Each row is specified by a small memory profile: storage type, context length, compression rule, read-out path, and recurrence structure when applicable. Mutual-exclusion constraints are enforced at the profile level; for example, the two State-Space instantiations cannot be enabled simultaneously, and length compression is kept separate from pure token weighting unless the row is explicitly marked as hybrid. This unified interface is what makes Sec. 6 a controlled comparison rather than a juxtaposition of independently tuned systems.

Table 1 Single-stage hyper-parameters shared by all variants. Only the columns marked (varies) are perturbed across variants.

Setting	Value
Backbone	Video DiT (per-frame VAE)
Resolution	352×640
Segment length T	81 frames
Context length K	<u>(varies)</u> – $\{1, 5, 20\}$, default 5
Memory module	<u>(varies)</u> – Context, Compression, Spatial, or State-Space
Optimizer	AdamW
Learning rate	5×10^{-5} (1×10^{-4} at $K=1$)
Per-device batch / grad accumulation	1 / 1
GPUs	8 (A100-80G)
Total steps	5 k
Timestep shift	15
Spike rejection threshold	15.0
Target-frame-only supervision	enabled
Flow noise shift	1.0
Relative-RT action encoding	enabled
Context placement	suffix
10% overlap-drop policy	enabled

4 Training and In-training Diagnostics

By fixing the backbone, optimizer, schedule, and inference path across variants (section 3.4), the training setup is a constant of the study: the only thing that varies across the rows of table 3 is the memory module. This is what lets the later differences be read as storage, compression, read-out, or recurrence effects rather than as hidden protocol changes. We describe that constant below, and then promote in-training replay sampling from a debugging convenience to a diagnostic signal.

4.1 Data

We train on the Context-as-Memory dataset [62], which contains real-world camera trajectories paired with per-frame poses and textual prompts. Each training sample is an 81-frame segment at 352×640 resolution, drawn from a long source video. For each sample, the data interface provides (i) the target segment $\mathbf{x}_{s:e}$, (ii) the corresponding 12-dim relative-RT camera trajectory, and (iii) a context buffer of $K-1$ historical frames retrieved from pre-computed field-of-view overlap labels keyed on the mid-frame of the segment.

Train/eval reference-frame alignment. A non-trivial implementation invariant is that the same notion of reference frame is used at training and evaluation time when retrieving context: training uses $r_{\text{train}} = (s+e)/2$, while in-domain evaluation uses $r_{\text{eval}} = N_{\text{frames}}/2$ on videos that always start at frame 0. This alignment is not a headline ablation, but we still treat it as a hard protocol invariant.

4.2 Single-stage Optimization

All variants in this study are trained in a single stage, of identical length, with the optimizer settings listed in Table 1. We deliberately avoid multi-stage curricula and stage-1 / stage-2 splits: in pilot experiments at this scale, we were unable to find a multi-stage schedule that consistently outperformed the single-stage one, and the simpler protocol makes the ablations easier to read.

4.3 Replay Sampling Methodology

A practical observation from pilot experiments is that the scalar flow-matching loss is a poor progress indicator for action-world-model training. Two runs with visually different boundary behavior can have nearly indistinguishable denoising-loss curves; conversely, a run with a misconfigured camera-action condition can look stable on the loss until a return trajectory exposes the error. We therefore use replay sampling as an



Figure 3 Replay progression on a fixed GT camera trajectory. The first row is the dataset trajectory; the remaining rows show generated replay frames from the Context $K=5$ and Context $K=20$ rows at matched time indices. This panel is a visual companion to the Replay PSNR/SSIM/LPIPS metrics, showing where background structure and object layout begin to drift across chunks.

in-training methodology, not as an informal debugging video. The replay protocol fixes the sample pool, the action construction, and the generation code path, so changes in the sampled videos can be attributed to the memory profile under training, and boundary continuity, identity preservation, and action-visual alignment become directly comparable across variants well before final evaluation.

Sample pool and segment length. Every variant in Sec. 6 is configured to sample, at fixed step intervals, videos built from the same held-out first frames. Each generated segment contains 81 frames, matching the training target length, and multi-segment replay concatenates several such segments without changing the memory interface between segments. This keeps replay close to the actual inference regime, rather than measuring a short single-clip proxy.

Camera-trajectory construction. Replay actions are produced by the same relative-RT constructor used in evaluation. For ground-truth replay, the constructor reads the dataset camera trajectory and converts each frame into a 12-dim relative RT vector. For return probes, it synthesizes symmetric yaw motions, including $180^\circ/360^\circ$ loops and the shorter 45° open-domain revisit used when no GT pose is available. The important invariant is that the trajectory is generated before the model sees the target frames, so replay remains a test of action conditioning and memory rather than a post-hoc alignment procedure.

Shared generation path. Sampling uses the same generation procedure as evaluation, with the same context retriever, RT-relative actions, and memory profile. The sampled videos are therefore a faithful preview of evaluation behavior, not a divergent path.

Loss and metric granularity. The training loss in Eq. 1 is computed only on target frames within the current segment; context frames are conditioning evidence, not supervision targets. Replay diagnostics are stored at both per-frame and per-chunk granularity. Per-frame PSNR/SSIM/LPIPS reveal where drift begins, while chunk means and retention ratios summarize whether quality decays as memory must carry information farther from the anchor frame.

Multi-segment sampling as an early memory metric. Multi-segment replay is a useful preliminary metric because it stresses the same failure mode that revisit evaluation later quantifies: the model must preserve scene evidence after the immediate visual context has been replaced by generated chunks. The boundary continuity, identity preservation, and action-visual alignment visible in these samples are qualitative evidence that we inspect before interpreting the full evaluation bundle.

4.4 Protocol Invariants

Every model is evaluated under the following hard-fail invariants: (i) context handling matches between training and inference; (ii) context noise levels match between training and inference; (iii) K matches between

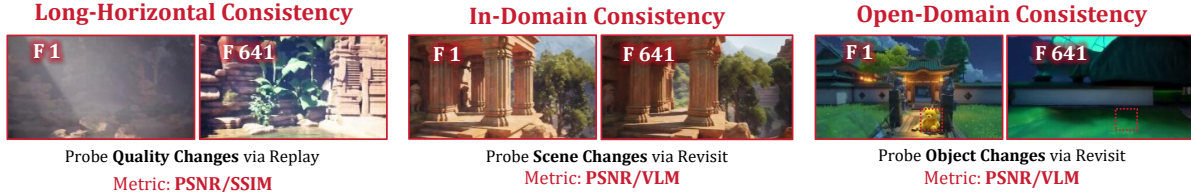


Figure 4 Evaluation taxonomy used in the study. Replay measures long-horizon image quality with PSNR/SSIM/LPIPS. The in-domain loop and open-domain return probe both report PSNR together with VLM scores, so pixel-level return quality and semantic scene/identity preservation are evaluated separately.

training and inference; (iv) target-frame-only supervision and the flow-noise shift remain fixed; (v) camera-action parameters are loaded before evaluation; and (vi) mutual exclusion between the two State-Space instantiations, and between Compression weighting and Compression length reduction, is enforced unless a row is explicitly marked as hybrid.

5 Evaluation

5.1 Overview

The central evaluation challenge is that action world models mix two problems that should not be collapsed into a single score. A model may fail to follow an observed camera trajectory, which is primarily an image-quality and control problem; or it may produce plausible local chunks but forget the scene when the camera returns, which is the memory problem studied in this paper. We therefore evaluate three camera-pose settings: **replay**, **in-domain loop closure**, and **open-domain fast memory probing**. Replay PSNR/SSIM/LPIPS plays a role similar to a low-cost pretraining monitor such as perplexity in language modeling: it is cheap and sensitive to basic reconstruction failures, but it is not a complete measure of long-horizon memory. We therefore report revisit PSNR, VLM verification, and qualitative inspection as a stacked evidence bundle rather than a single scalar.

5.2 Evaluation Samples

In-domain. The in-domain split uses dataset samples that come with first frames, metadata prompts, GT camera trajectories, and GT images. This support lets us report PSNR for loop-closure outputs and use VLM scoring for fine-grained scene and identity details.

Open-domain probes. The open-domain half uses first frames from a held-out revisit-probe set that is not used during training. These samples have no pixel-level GT trajectory, so we avoid treating open-domain pixel differences as geometric truth, and instead use short controlled memory probes scored by VLM verification.

Three evaluation branches. We use three named pose settings, each answering a different question. **Replay** follows the dataset camera trajectory and is used for long-horizon image-quality evaluation. This branch verifies that the model can synthesize frames under observed motion, and provides the PSNR/SSIM/LPIPS curves we use as a low-cost pretraining monitor. **In-domain random-motion revisit** uses the reported 180° return loop on dataset-backed samples, with longer 360° loops retained only as stress probes when explicitly stated. Because the outgoing and returning views are pixel-aligned, this branch tests whether memory remains decoupled from action control when the camera comes back to a known scene. **Open-domain 45° revisit** uses a shorter return probe on held-out first frames. The smaller angle is a deliberate trade-off: at current open-domain generation quality, longer unconstrained loops are unstable, while a controlled 45° revisit still exposes object and scene consistency failures.

5.3 Long-Horizon Replay Quality

The replay protocol drives the model with the dataset camera trajectory for multiple chunks and compares the generated frames to ground truth. It answers a narrow question: can the model synthesize high-quality

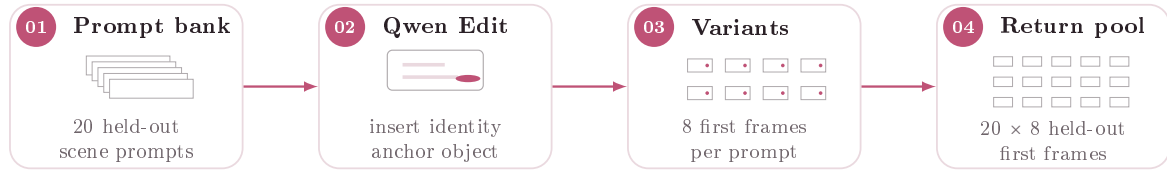


Figure 5 Qwen Edit construction of open-domain first frames. Each held-out environment prompt is edited into identity-anchored first frames, which seed the open-domain return probe.

frames while following observed camera motion? It is intentionally not used as the primary memory metric. A model with strong replay quality may still forget the scene once the trajectory becomes a return loop.

Metrics. Replay PSNR, SSIM, and LPIPS are the headline long-horizon metrics, because this branch focuses on image fidelity under GT camera motion. Their role is analogous to perplexity during language-model pretraining: they are not a semantic success criterion, but they are inexpensive, continuous, and useful for plotting whether basic reconstruction ability improves or degrades. We do not use distributional video scores as the main replay metrics, since replay is GT-aligned and the immediate goal is frame-level consistency rather than distributional video ranking. The export also stores per-chunk PSNR, SSIM, LPIPS, and retention ratios relative to the first generated chunk; these diagnose progressive degradation, while the tables report the mean over three chunks.

5.4 Static Consistency

Static consistency is the memory-sensitive branch. Both in-domain and open-domain probes report a paired evidence set: PSNR for visual return quality and a VLM score for semantic scene/identity preservation.

In-domain 180° loop. For in-domain samples, the 180° return loop gives a GT-backed revisit protocol. We report PSNR by matching frames at mirrored positions along the outgoing and returning trajectory, and averaging the pairwise scores. This symmetric mean directly asks whether the model returns to a visually consistent view rather than drifting to a different scene. We additionally use a VLM score to verify fine-grained details that pixel metrics may miss, such as object identity, local layout, and salient texture consistency.

Open-domain 45° memory probe. For open-domain samples, no GT trajectory or pixel reference is available. The primary protocol is therefore a compact left/right return probe: the camera turns 45° in one direction and then returns by 45°. This short loop is a deliberately practical choice. In our current setting, longer open-domain loops often fail for reasons that mix general image synthesis with memory, while the compact return probe still reveals whether memory improves the returned scene.

Open-domain first-frame generation. The open-domain pool is generated with Qwen Edit rather than sampled from the training set. For each held-out game-style environment prompt, we ask the edit model to insert a distinctive object while preserving the surrounding scene. Using 20 prompts and 8 edited variants per prompt yields a 20×8 first-frame pool whose objects are easy to name, localize, and revisit.

At evaluation time, each edited first frame seeds the 45° leave/return probe. We report a visual-return PSNR proxy and a Qwen3-VL-30B-A3B semantic score for SCENE and SPECIAL IDENTITY. The VLM prompt weights object appearance and presence more heavily than background plausibility:

$$0.45 s_{\text{appearance}} + 0.25 s_{\text{presence}} + 0.20 s_{\text{view}} + 0.10 s_{\text{scene}}, \quad (6)$$

with the $[0, 5]$ weighted sum rescaled to $[0, 100]$.

5.5 Robustness of the VLM Judge

Because the open-domain branch relies on a single VLM judge, we run a sanity check on whether the choice of judge changes our conclusions. The probe is a constrained recognition task: the judge compares the source frame with revisit-tail frames and decides whether the salient object, viewpoint, and scene are preserved. We re-score a stratified subset of open-domain cases with Qwen3-VL-30B-A3B, two stronger closed-source judges,



Figure 6 Open-domain revisit source panel. The 2x4 grid shows representative first-frame sources from the Qwen-edit open-domain pool. The full pool is generated by editing distinctive toy-like objects into held-out game-style environments; a practical construction uses 20 scene prompts and 8 edited variants per prompt. The displayed examples cover temple, urban, ruin, rooftop, courtyard, street, and beach settings. These objects serve as deliberately simple identity anchors for the VLM judge: after the camera leaves and returns, the judge scores whether the same object appearance, subject presence, camera viewpoint, and background scene are preserved. No annotations are drawn inside the panel, so the figure remains a clean visual inventory of the evaluation cases.

Table 2 Cross-judge agreement for open-domain scoring. Mean scores over a shared open-domain subset. Δ is the mean absolute difference from Qwen3-VL-30B-A3B, and ρ is Pearson correlation with Qwen3-VL.

Judge	Mean score	Δ vs. Qwen	ρ vs. Qwen
Qwen3-VL-30B-A3B	19.6 $\pm$ 13.9	–	1.00
Claude Opus 4.6	20.7 $\pm$ 11.1	3.1	0.93
GPT-5.5	21.1 $\pm$ 11.4	2.7	0.94
Human	20.9 $\pm$ 11.9	1.3	0.96

and a human anchor, all receiving the same images and prompt (table 2). The alternative judges stay within a few points of Qwen3-VL on average, remain correlated above 0.90, and preserve the relative reading of the subset. Changing the judge therefore adds at most a small offset rather than reordering the memory profiles, so we keep the open-source judge as the default scorer and treat the closed-source judges as cross-validation only.

6 Empirical Analysis

The ablations are organized around the failure mode that motivates the paper: a model can generate plausible local video while still failing to remember the world at revisit time. Each sub-section isolates one axis of the memory design space and reads the resulting numbers through the same three-branch evidence stack established in section 5. The headline table compares four memory families: Context, Compression, Spatial, and State-Space. The mechanism tables then isolate three choices: Spatial read-out, Compression design, and State-Space recurrence. A final Context table reports the benefit of longer raw history. Unless stated otherwise, revisit numbers are the current material snapshot from eight cases per row: in-domain uses the 180° return loop, while open-domain uses the 45° return probe. Replay columns come from the GT-trajectory replay export and report mean PSNR/SSIM/LPIPS over three chunks.

How to read the tables. The tables should be read horizontally before they are read vertically. Each row reports three different views of the same model. Replay PSNR/SSIM/LPIPS measures whether the row can follow an observed trajectory with acceptable low-level fidelity; in-domain revisit asks whether the model can return to a known scene under a controlled loop; and open-domain VLM asks whether the same memory mechanism preserves a salient object and scene when no pixel-aligned reference is available. A row that wins on one column but fails on another is therefore not an outlier to be explained away. It is evidence that the memory mechanism is solving only part of the action-world-model problem, and the particular column that

Table 3 Representative memory-family comparison. R-P/R-S/R-L denote Replay PSNR/SSIM/LPIPS; ID-P/ID-S/ID-L denote in-domain closure PSNR/SSIM/LPIPS; O-V denotes open-domain VLM.

	R-P↑	R-S↑	R-L↓	ID-P↑	ID-S↑	ID-L↓	O-V↑
I2V baseline	10.03	0.398	0.534	10.32	0.291	0.643	12.25
Spatial Memory baseline	13.60	0.411	0.554	10.04	0.260	0.617	6.00
Compression weight-only	8.65	0.174	0.752	10.67	0.154	0.645	22.38
State-Space (legacy hybrid)	12.69	0.344	0.581	12.23	0.298	0.584	34.75
State-Space (block-wise)	9.59	0.282	0.698	11.95	0.280	0.535	69.00
Context learning, $K=5$	11.92	0.408	0.501	10.72	0.307	0.596	50.75
Context learning, $K=20$	12.54	0.449	0.496	11.07	0.359	0.543	58.63

breaks tells us which part. This is the diagnostic logic we will use throughout the section.

6.1 Memory Families under Matched Evaluation

The headline result immediately shows why replay alone is insufficient. Different low-level metrics select different winners: Spatial Memory is strongest on replay PSNR, long raw Context is strongest on replay SSIM/LPIPS, and block-wise State-Space remains strongest on in-domain LPIPS and open-domain VLM. This inversion means that memory quality is not the same thing as image quality under GT motion. Replay rewards the ability to synthesize frames that match a dataset trajectory, while open-domain revisit rewards the ability to carry object identity and scene layout through a generated excursion. The Context rows occupy a useful middle ground: increasing raw history improves semantic return much more clearly than it improves the full replay or in-domain image-metric bundle.

The Spatial-State-Space inversion is the central finding of the table. Spatial Memory wins replay PSNR and remains competitive on the replay image bundle, but collapses on open-domain VLM; block-wise State-Space loses the replay bundle and yet wins open-domain VLM by a wide margin. Two rows on the same backbone, optimizer, sampler, and evaluation pipeline thus arrive at opposite conclusions about which is the better world model, depending only on which column one reads. This is exactly the kind of evidence that a single-metric report would suppress. It also says something stronger: replay PSNR cannot be treated as a low-cost proxy for revisit quality. A model that reconstructs GT trajectories at sub-pixel cost can still fail to preserve a single salient object across a 45° excursion, while a model that produces less faithful pixels can nonetheless carry the same scene through it. Replay measures how well a model follows; revisit measures how well a model remembers; these are different capabilities, and they are not monotonically aligned.

The in-domain column tells a third, intermediate story. Several rows cluster in the same in-domain VLM range, and the best in-domain PSNR comes from the legacy State-Space variant, while the best in-domain LPIPS comes from block-wise recurrence. This should not be read as a contradiction. In-domain return has pixel-aligned views and familiar data statistics, so a model with strong local reconstruction can partly succeed simply by tracking the trajectory back to a visually similar view. Once the open-domain probe removes that support, the picture changes: object preservation becomes the dominant criterion, and the block-wise row separates from the legacy variant. Reading the two columns together is therefore informative in a way that neither column is on its own. It exposes which rows were quietly relying on in-distribution texture statistics and which were carrying scene evidence in a form robust enough to survive shift.

What about Context? The Context rows are easy to under-rate because they contain no new machinery. Yet $K=20$ posts the second-best in-domain VLM (29.40) and a solid open-domain VLM (58.63), within striking distance of block-wise State-Space, while staying mid-pack on Replay. In a field that prefers learned bottlenecks to brute capacity, this is a counter-result: raw retrieval, with no learned compression and no specialized read-out, beats every compact memory mechanism in the matrix on open-domain semantic return except block-wise recurrence. We will see in [section 6.5](#) that the gain is not linear in K either: most of the open-domain jump happens by $K=5$, with $K=20$ supplying further refinement rather than a step change.

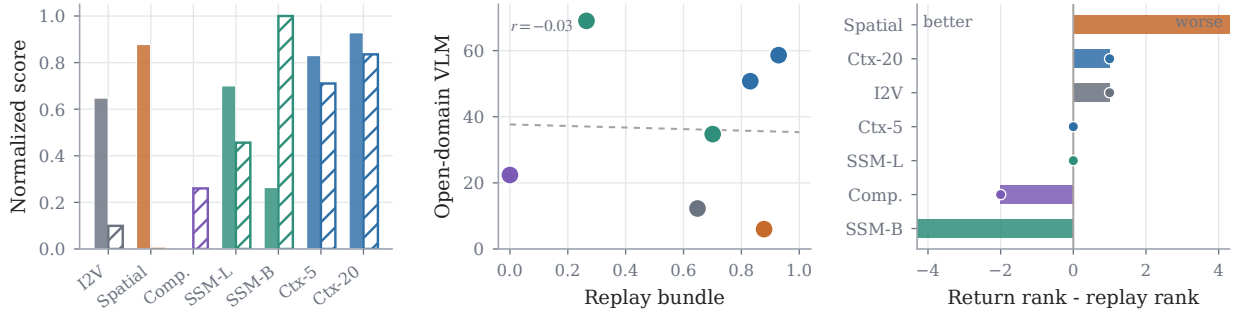


Figure 7 Replay is a health signal, not the final memory score. The three panels compare normalized replay and return scores, plot replay bundle against open-domain VLM, and show the rank shift from replay to return. In the first panel, solid bars are replay and hatched bars are return; in the third panel, negative values indicate a better rank under return. Colors denote Context, Spatial, Compression, State-Space, and I2V.

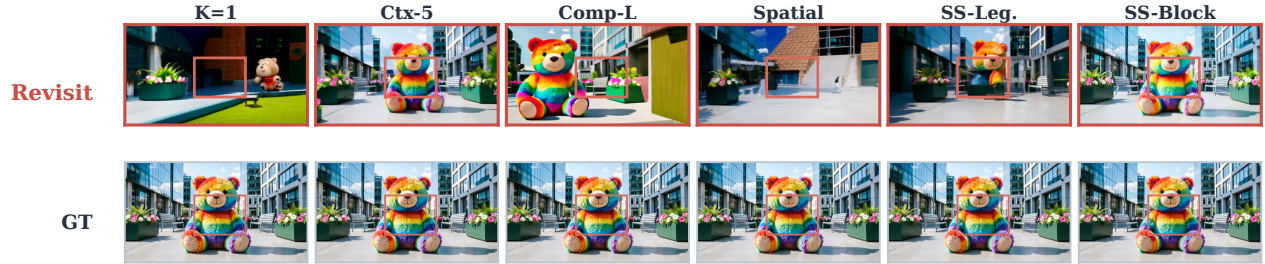


Figure 8 Representative open-domain revisit case. Columns are matched qualitative rows, except for Context $K=20$. The emphasized top row shows the matched revisit frame for each memory family, and the bottom row repeats the GT/reference view for comparison. The light box marks the same salient-object region across methods, so identity drift is visible before consulting any scalar score.

Treating the headline table as a metric bundle. For the reasons above, the paper treats this table not as a ranking but as a metric bundle. Replay filters out broken generators that cannot follow a trajectory at all; in-domain return checks controlled loop closure under familiar statistics; open-domain VLM provides the strongest evidence of whether the memory survives distribution shift. The most informative rows are precisely the ones where these three views disagree: Spatial Memory and block-wise State-Space disagree by sixty-three VLM points on open-domain return alone. The remainder of the section explains why.

Finding 1: Open-domain return is the most discriminative memory stress test. Replay and in-domain return are necessary filters, but the open-domain VLM column is where memory families separate most clearly.

Why the alignment plot matters. figure 7 turns the table into a diagnostic question. Replay is not meaningless: it catches whether a row can synthesize coherent frames under GT camera motion, so it is a dense and cheap health signal for intermediate model selection. But the same plot also shows why mean replay quality cannot be promoted into the final memory score. The strongest replay rows can still fail open-domain return, while rows with weaker replay fidelity can carry scene evidence through the excursion. Replay tells us whether the generator is healthy enough to trust; revisit tells us whether the world was remembered.

6.2 Spatial Storage and Read-out

Setup. Spatial Memory asks whether a compact scene summary can replace raw temporal context. The ablation separates writing from reading: storing tokens but never injecting them is a lower bound; appending

Table 4 Spatial Memory read-out ablation. R-P/R-S/R-L denote Replay PSNR/SSIM/LPIPS; ID-P/ID-S/ID-L denote in-domain closure PSNR/SSIM/LPIPS; O-V denotes open-domain VLM.

	Read-out	R-P↑	R-S↑	R-L↓	ID-P↑	ID-S↑	ID-L↓	O-V↑
Spatial baseline	default	13.60	0.411	0.554	10.04	0.260	0.617	6.00
Spatial inject-none	none	14.66	0.417	0.541	10.38	0.309	0.631	15.50
Spatial concat-text	text KV concat	13.44	0.412	0.572	9.35	0.272	0.614	10.25
Spatial cross-attn RO	separate cross-attn	13.52	0.409	0.557	8.49	0.232	0.664	17.12

them to text cross-attention and reading them with a dedicated cross-attention block are the active variants. This distinction matters because a compact store can look useful for two different reasons. It may help by providing the generator with a stable scene representation, or it may simply regularize the context pathway and improve local reconstruction without carrying the object evidence needed for revisit. The read-out rows are designed to separate these cases: **withheld read-out** tests whether storage alone changes the generator, **text KV concatenation** tests whether the existing cross-attention pathway can consume the summary, and **dedicated read-out** tests whether the summary needs its own access path.

The replay numbers are encouraging but also cautionary. Spatial inject-none obtains the strongest replay PSNR/SSIM/LPIPS bundle among the spatial rows, even though it withholds the stored tokens from the generator. The open-domain VLM column shows that stronger read-out can help: dedicated cross-attention reaches 17.12, above the default spatial baseline at 6.00. Yet the same column also shows that the family is still weak as semantic memory. Even the best read-out row remains far below raw Context $K=20$ and block-wise State-Space in [table 3](#), and inject-none remains surprisingly competitive at 15.50. The spatial store is therefore not simply missing a read-out path; its compact representation also appears to discard object evidence needed for return.

Why does withheld read-out improve Replay? The most striking number in [table 4](#) is the inject-none row: 14.66 Replay PSNR, 0.417 SSIM, and 0.541 LPIPS, all better than the active read-out variants. Withholding the stored tokens makes the network strictly less expressive at inference, yet it makes replay look strictly better. The simplest explanation consistent with the rest of the matrix is that the spatial summary is not, at this scale, behaving as a memory at all: it is behaving as an additional training-time conditioning channel that regularises the network and quietens the per-chunk reconstruction error. Once we remove the tokens at inference, the network reverts to the cleaner I2V-style prediction it has implicitly been prepared for, and replay PSNR ticks up. The lesson is that replay improvements from a memory module do not, on their own, certify that the module carries memory.

Storage cannot be evaluated without read-out. The read-out rows span a useful gradient of access strength. Withheld read-out is a hard zero: tokens exist but are unreachable. Text-KV concatenation is a soft access path: the spatial tokens are dumped into the same cross-attention slots that already serve the text prompt. Dedicated cross-attention read-out is a harder access path: the spatial tokens get their own query mechanism and a zero-initialised gate that the optimizer can open. The measured VLM scores indicate that access matters, but not enough. Dedicated read-out is best among the spatial rows, while text-KV concat improves only mildly over the default baseline. Since all three read-out variants remain far below the strongest context and State-Space rows, the failure is shared between access and storage bandwidth, rather than being only missing bookkeeping in the read-out module.

Why Spatial loses open-domain VLM by so much. A finer reading of the score helps explain the magnitude. The open-domain probe is constructed so that the salient object is distinctive in shape, colour, and category; in effect, it is chosen to be a high-entropy identity anchor. A $G \times G$ grid summary is a low-bandwidth representation of the scene; it can encode where things roughly are, but it has limited capacity to encode which thing. When the camera leaves and returns, the generator must place the same object back into the frame, and a generic spatial code under-determines that decision. Replay does not expose this gap because the GT trajectory keeps the object visible; revisit exposes it because the object must be regenerated from memory alone. The Spatial family is therefore not weak in general; it is weak exactly where compact summaries should be weak, which is where compact memory bottlenecks are most attractive to use.

Table 5 Compression ablation. R-P/R-S/R-L denote Replay PSNR/SSIM/LPIPS; ID-P/ID-S/ID-L denote in-domain closure PSNR/SSIM/LPIPS; O-V denotes open-domain VLM.

	Mechanism	R-P↑	R-S↑	R-L↓	ID-P↑	ID-S↑	ID-L↓	O-V↑
Weight-only baseline	token weighting	8.65	0.174	0.752	10.67	0.154	0.645	22.38
Length $r=2$	temporal length	7.84	0.162	0.746	8.82	0.183	0.654	24.00
Length $r=4$	temporal length	9.88	0.183	0.714	9.53	0.163	0.617	43.25
Hybrid $r=2$ + weight	both	9.10	0.181	0.734	7.87	0.277	0.695	8.75
Hybrid $r=4$ + weight	both	9.63	0.215	0.730	8.81	0.156	0.625	9.00

Finding 2: Spatial summaries are efficient but not yet reliable as semantic memory. The open-domain gap points to read-out design and to the bandwidth of the storage grid as the joint limiting factor; replay efficiency alone cannot tell which of the two is binding.

6.3 Compression and Temporal Resolution

Setup. The Compression family asks whether memory should be compacted by importance or by time. Token weighting preserves the number of historical observations but changes their relative influence; length compression discards temporal resolution. The hybrid row tests whether the two operations are complementary or redundant. These variants are often grouped together as “compression”, but they make different assumptions about what history contains. Token weighting assumes that all temporal positions can remain available while their importance varies with age or relevance. Temporal length reduction assumes that neighboring observations can be pooled without destroying the memory signal. The hybrid row is the strictest test: it asks whether the model can first reduce temporal support, and then still learn a useful importance profile over the reduced support.

The table shows that compression has no monotonic ordering by apparent capacity. Weight-only compression is the most conservative operation because it keeps every temporal position, yet it is not the best open-domain semantic row. Length $r=4$ outperforms weight-only on open-domain VLM, indicating that a shorter but cleaner context can sometimes be preferable to a full-length context with learned weights. However, the hybrid rows perform poorly, especially on open-domain VLM. The likely explanation is that the two operations remove different kinds of signal: pooling reduces temporal evidence before the model can decide which observations matter, and weighting then operates on an already coarsened sequence. When the return probe depends on a particular object or view, this order can erase the evidence that weighting would otherwise have preserved.

Compression is not a monotone capacity dial. Naive intuition says that “less compression” should mean “more memory”. The numbers in [table 5](#) actively disagree. Weight-only compression preserves every temporal position ($r=1$ in length terms) and yet posts an open-domain VLM of only 22.38. Length $r=4$ throws away three out of every four temporal positions, and reaches 43.25 on the same metric. The model that sees less history does better on the harder column. This is counter-intuitive only if one assumes that all temporal frames carry equal information; once we drop that assumption, the result becomes mechanistic. A length-pooled context concentrates the remaining capacity on fewer, cleaner temporal slots, which makes the per-frame signal easier for the generator to read; a weight-only context preserves every frame but spreads the model’s attention thinly across them, and the attention budget for any specific frame stays low. The lesson is structural: how the context is shaped can matter more than how much of it survives.

Why hybrid compression backfires. The hybrid rows are the most diagnostic numbers in the table. They combine the two operations one would naively assume to be complementary, and they perform worse than either operation alone (8.75 and 9.00 on open-domain VLM, well below both pure weighting and pure length pooling). This is not a small effect: hybrid compression discards three-quarters of the open-domain return signal that pure length pooling preserves. The order of operations explains why. Length pooling acts first, on the raw context. It collapses neighbouring frames into a single pooled slot without consulting any importance

Table 6 State-Space memory comparison. R-P/R-S/R-L denote Replay PSNR/SSIM/LPIPS; ID-P/ID-S/ID-L denote in-domain closure PSNR/SSIM/LPIPS; O-V denotes open-domain VLM.

	Structure	R-P↑	R-S↑	R-L↓	ID-P↑	ID-S↑	ID-L↓	O-V↑
State-Space (legacy hybrid)	legacy hybrid	12.69	0.344	0.581	12.23	0.298	0.584	34.75
State-Space (block-wise)	paper-aligned recurrence	9.59	0.282	0.698	11.95	0.280	0.535	69.00

signal, so any frame that happens to share a window with a forgettable neighbour is averaged out before the model can object. Weighting then learns importance on top of this coarsened sequence, where the high-value frames are no longer individually addressable. The two operations do not compose like filters; they compose like a destructive quantization followed by a re-encoding of the resulting noise. [figure 8](#) makes the failure mode visible: hybrid rows tend to preserve the existence of an object after return, but not the identity of the object.

Cost-control, not capacity substitution. The replay columns reinforce the same interpretation. Length $r=4$ improves the Replay PSNR/LPIPS bundle over weight-only, but the absolute values remain below the strongest Context and Spatial rows. Compression therefore appears useful as a cost-control mechanism, a way to trade some replay quality for cheaper steps, not as a free substitute for historical capacity. The right reading of [table 5](#) is thus less about which compression operator wins and more about what compression is. In the current matrix, it is a budget-management tool that has not yet learned what to budget for. A stronger compression memory would need an explicit objective for what to retain: an object-aware pooling rule, a learned criterion tied to revisit success, or a mechanism that preserves a small set of high-value frames while compressing only the redundant background evidence around them. The current operators do none of these things, and the numbers reflect that absence.

Finding 3: Compression mechanisms do not compose monotonically. Length reduction can help, but combining it with attention weighting can discard the very temporal evidence needed for revisit. Effective compressed memory needs an explicit objective for what to retain, not just a smaller token budget.

6.4 Implicit State-Space Memory

Setup. State-Space memory asks whether an implicit recurrent state can carry the scene history that explicit context provides. We report two instantiations as one family: a legacy VideoSSM hybrid, and a paper-aligned block-wise recurrence. The point is not to crown a new State-Space variant, but to test whether implicit memory is a credible answer to revisit consistency. The comparison is deliberately narrow. Both rows use the same evaluation bundle and are treated as instantiations of an implicit-memory family, but they differ in where and how recurrence is structured. The legacy hybrid behaves like a lightweight recurrent supplement to the video backbone; the block-wise row makes recurrence a more regular part of the temporal computation. If both rows behaved similarly, the main conclusion would be that implicit memory is largely a family-level choice. Their separation instead suggests that recurrence design matters as much as the decision to use recurrence at all.

This is the clearest example of metric separation in the paper. Legacy hybrid recurrence has the best in-domain PSNR and strong replay PSNR, but block-wise recurrence dominates open-domain VLM. The likely reason is that in-domain return still offers familiar textures, motions, and scene statistics, so a model with strong local reconstruction can look good even when its state is not robust under distribution shift. Open-domain identity preservation is less forgiving. The model must carry a compact description of the salient object and environment through the generated excursion, and the block-wise recurrence appears better suited to this role.

This does not mean that State-Space memory is universally superior. Its replay PSNR is not the highest, and the family still needs comparison at longer open-domain horizons. What the current evidence supports

is more specific: a recurrent state can be a real memory mechanism for revisit, provided that its structure is strong enough to preserve scene evidence rather than merely smoothing the temporal computation. This motivates future SSM variants that expose the state to explicit object- or view-level supervision during replay and return probes.

The largest single jump in the paper. The two State-Space rows separate sharply on open-domain VLM using the same backbone, optimizer, sampler, and data, while their replay metrics move in the opposite direction and their in-domain PSNRs stay close. We are not aware of another row pair in the matrix that opens a comparable gap on open-domain VLM with so few controlled differences. The headline of [section 6.4](#) is therefore that the choice of recurrence structure, including where the state is updated, how often it is read, and how it is gated, is at least as consequential for revisit consistency as the older choice between explicit and implicit memory. Treating “State-Space memory” as a single design choice obscures the biggest variance the matrix has shown so far.

Why does block-wise recurrence preserve more under shift? A useful contrast is the legacy hybrid, which behaves like a lightweight recurrent supplement: it adds a recurrent gate around an otherwise standard temporal attention path. That gate is sufficient to smooth in-distribution motion and is sufficient to post a strong in-domain PSNR, but its state is shallow in the sense that any single block can largely bypass it. Block-wise recurrence, in contrast, makes the recurrent step a regular, non-bypassable part of the temporal computation: every n -th DiT block reads the carried state, updates it, and writes it back. The state thus accumulates evidence across the chunk boundary in a way that the legacy variant cannot, and it is this carried evidence that becomes load-bearing once the camera leaves the visible support. From the generator’s point of view, the block-wise state is the only place where information about the salient object can survive between segments without an explicit retrieval step.

Why does block-wise lose Replay? The same structural choice has a cost. A non-bypassable recurrent state forces the network to commit some representational capacity to summarising the past, and that capacity is not available for fitting the local trajectory pixel-by-pixel. The result is the 9.59 Replay PSNR and 0.698 Replay LPIPS, which on their own would put block-wise recurrence near the bottom of the headline table. Reading Replay alongside open-domain VLM is therefore essential here in a way that it is not for any other row. Block-wise recurrence is precisely the kind of design that a single-metric report would dismiss.

Implication: implicit memory is a real mechanism, not a smoother. The most robust statement we can make from [table 6](#) is narrow: a recurrent state can act as a genuine memory mechanism for revisit, provided that its structure is strong enough to preserve scene evidence rather than merely smooth the temporal computation. This is a stronger claim than “recurrence helps”. It rules out the lightweight-gate interpretation of state-space memory in this regime, and it sets a structural minimum: the state has to be carried through a non-trivial fraction of the network’s depth before it begins to behave like memory. Future SSM variants that expose this state to explicit object- or view-level supervision during replay and return probes are a natural next step, and the gap between 34.75 and 69.00 suggests there is significant room above the current ceiling.

Finding 4: Block-wise recurrence is the strongest current open-domain memory bias. The gain over legacy recurrence shows that the structure of implicit memory matters.

6.5 Raw Context Capacity

Setup. Context learning asks how much raw history is enough when the memory is not compressed into a special module. Here the I2V baseline contains only the current anchor frame; $K=5$ and $K=20$ add retrieved historical observations. This is the simplest ablation in the section, and also the most important reference point. Raw context has no learned compression bottleneck and no special read-out module, so it answers a direct capacity question: how much does the generator benefit when it can see more historical observations under the same retrieval and action-conditioning interface? Any compact memory should be compared

Table 7 Context length ablation. R-P/R-S/R-L denote Replay PSNR/SSIM/LPIPS; ID-P/ID-S/ID-L denote in-domain closure PSNR/SSIM/LPIPS; O-V denotes open-domain VLM.

	K	R-P $\uparrow$	R-S $\uparrow$	R-L $\downarrow$	ID-P $\uparrow$	ID-S $\uparrow$	ID-L $\downarrow$	O-V $\uparrow$
I2V baseline	1	10.03	0.398	0.534	10.32	0.291	0.643	12.25
Context learning, $K=5$	5	11.92	0.408	0.501	10.72	0.307	0.596	50.75
Context learning, $K=20$	20	12.54	0.449	0.496	11.07	0.359	0.543	58.63

against this trend, because a method that only beats the anchor-only baseline may still be weaker than ordinary uncompressed history.

Both replay and return improve with longer raw context, but the open-domain VLM trend is much steeper than the replay trend. Replay PSNR, SSIM, and LPIPS improve from the anchor-only baseline to $K=20$, while open-domain VLM makes most of its gain by $K=5$ and continues to improve at $K=20$. This asymmetry is meaningful: raw history improves the GT-trajectory image bundle, but it buys much more on the semantic return axis, where the model must restore an object and scene after leaving the view. In other words, raw context is not merely extra conditioning; it is a memory-capacity baseline.

The result also calibrates the rest of the study. Spatial Memory and compression are attractive only if they preserve enough of this raw capacity at lower cost, while State-Space memory is attractive if it matches or exceeds raw context in robustness under distribution shift. The current table suggests that raw context remains difficult to beat as a simple solution. The best compact or implicit memory mechanisms should therefore be evaluated against Context $K=20$, not only against the I2V baseline.

The capacity asymmetry between Replay and revisit. The single most informative pattern in [table 7](#) is the slope mismatch between the replay image bundle and open-domain return. Replay PSNR moves by 2.51 points from $K=1$ to $K=20$, while open-domain VLM moves by 46.38 points over the same range. Raw history does improve pixel-level reconstruction, but its larger effect is on the ability to put the same object back into the frame after the camera has left it. This is the cleanest evidence in the paper that revisit consistency is a different capability from trajectory following, and that the two scale with the available history at different rates. Any compact memory whose argument rests on “matching Replay quality at a fraction of the tokens” has, by this measurement, only matched the cheaper of the two quantities.

The gain is sublinear in K . Most of the open-domain VLM jump happens by $K=5$ (+38.50 points over the I2V baseline), with $K=20$ adding a further 7.88 points. The marginal value of an extra retrieved frame is therefore high near the cold-start regime and decays quickly. Two consequences follow. First, the I2V baseline is severely under-resourced: it is not a calibrated reference point for what a generation model can do, because moving from $K=1$ to $K=5$ alone roughly quadruples the open-domain VLM score on the same backbone. Any memory method that primarily reports “beats I2V” should be re-read with this factor in mind. Second, the asymptote of raw context appears to be near, not at, $K=20$: the cost grows linearly with K , the benefit grows sublinearly, and the curve is already flattening. This is exactly where compressed memories should help. The fact that they do not in the current matrix is the strongest invitation for future work that the paper makes.

Raw context as a calibration tool. The right way to read $K=20$ Context is therefore not as a candidate winner of the matrix, but as a calibration tool. It tells us what a generator can do when it is given uncompressed evidence under the same retrieval and action-conditioning interface as the specialized methods. Any compact or implicit memory should be evaluated against [this](#) reference point, not against the I2V floor. Under that more demanding bar, the current matrix shows exactly one mechanism, block-wise State-Space, that exceeds Context $K=20$ on open-domain VLM, and most of the others fall short of even $K=5$. This runs against the field’s default direction of travel toward more elaborate compressed memories.

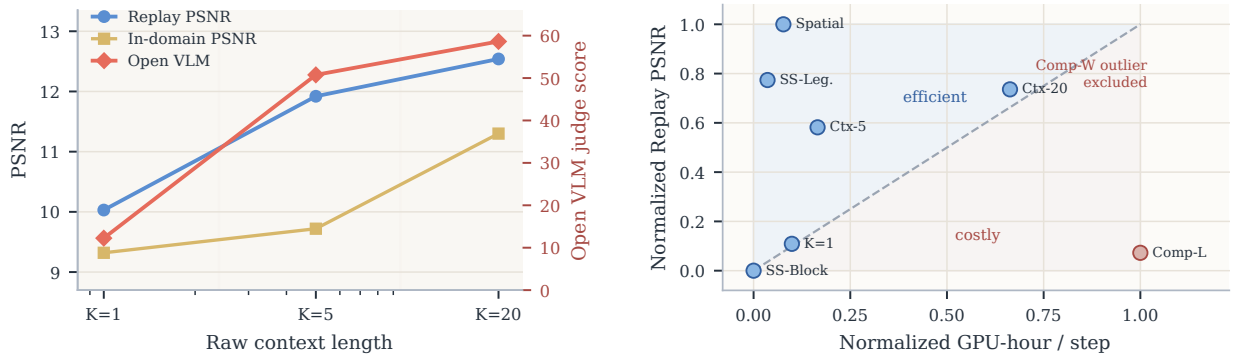


Figure 9 Context scaling and training efficiency. Left: the I2V baseline is anchor-only; increasing raw context improves open-domain semantic return, while the PSNR, SSIM, and LPIPS replay curves separate from that trend. Right: the efficiency grid compares normalized Replay PSNR against normalized GPU-hour per step, after excluding the Compression weight-only outlier from the plotted range. Points above the diagonal deliver more replay quality per unit step cost.

Finding 5: Raw context is the capacity baseline, and the gain it buys is overwhelmingly on revisit rather than on Replay. Compact and implicit memories should be judged by how much of the $K=20$ open-domain benefit they retain, not by how much they beat I2V.

6.6 Quality–Time Trade-off

Setup. The headline table ranks quality after training, but a practical memory baseline must also justify its training cost. We parse the latest training logs for each representative row, convert the average step time to GPU-hour per optimization step, and join it with the final Replay PSNR reported in `materials/training_time_by_baseline.csv`. The scatter below normalizes both axes: smaller x means cheaper steps, larger y means stronger replay fidelity, and the diagonal marks equal normalized quality and cost. The efficiency panel should be read as a diagnostic rather than as a replacement for the evaluation bundle. It measures how much replay quality is obtained per unit of step cost, which is useful for deciding which memory rows are practical to scale. However, the axis is replay PSNR, not open-domain semantic return: a method can look efficient because it reconstructs GT trajectories cheaply while still failing to preserve object identity after a return. Conversely, a method with weaker replay efficiency can still be scientifically important if it is the only one that preserves open-domain memory.

This distinction explains why the final conclusion is not a simple efficiency ranking. Spatial Memory is attractive in a replay-efficiency view because it delivers strong low-level reconstruction at a compact cost, but Finding 2 shows that this efficiency does not automatically transfer to open-domain identity preservation. Block-wise State-Space has the opposite profile: it does not dominate replay PSNR, yet it is the most reliable semantic-return mechanism in the open-domain probe. Raw Context is the third corner of the trade-off: it provides capacity and strong semantic return, but its cost grows with the amount of retrieved history. The practical memory-design problem is therefore a three-way trade-off among replay fidelity, step-time efficiency, and revisit generalization.

Three corners, three different design recipes. It is worth stating the three corners of this trade-off explicitly because they map onto three different downstream uses of an action world model. (i) If the application is high-fidelity replay of an observed trajectory at the lowest possible step cost, such as telemetry-driven re-rendering of a recorded camera path, the replay-efficiency corner argues for compact spatial summaries. (ii) If the application is interactive camera control over a scene that the camera will leave and re-enter many times, the canonical world-model use case, the open-domain VLM corner argues for block-wise state-space recurrence, and the rest of the matrix is at best a second choice. (iii) If neither cost nor open-domain robust-

ness is the binding constraint, raw context at $K=20$ is the safest default, because it dominates the in-domain VLM column and is strictly within reach of block-wise State-Space on open-domain return. [figure 9](#) should be read with this map in mind: points above the diagonal are good for replay alone, and they are not necessarily good for the use case that motivated the world-model framing.

Why a single efficiency number would mislead. Suppose one collapsed the matrix into a single “quality per GPU-hour” scalar by averaging Replay PSNR with a normalised cost. Spatial Memory would win, block-wise State-Space would look unappealing, and the headline of the paper would be that compact summaries are the practical answer. Yet the open-domain VLM column disagrees by a factor of more than $11\times$ between the two (69.00 vs. 6.00). The single-number ranking would not be wrong about the cost; it would be wrong about what the cost is buying. This is the structural reason the paper insists on reporting the metric bundle rather than collapsing it.

Finding 6: Efficiency must be evaluated together with revisit generality. Optimizing the replay-efficiency panel alone would select a different memory than optimizing for a revisitable world. The corner of the trade-off that matters depends on whether the downstream task is trajectory replay or open-ended revisit, and the right design recipe differs between the two.

6.7 Cross-cutting Observations

Read together, the sub-sections support three statements that no single row establishes on its own.

(A) Replay quality and revisit quality are different axes that routinely disagree. Spatial Memory dominates Replay and bottoms out on open-domain VLM; block-wise State-Space does the reverse; length $r=4$ is worse than its hybrid counterpart on Replay but far better on revisit. Across the matrix the rank correlation between Replay PSNR and open-domain VLM is weak and sometimes inverted within a family, because the two metrics measure two sub-problems: following an action sequence and remembering a world. A memory paper that does not separate these axes cannot tell whether its mechanism helps generation or memory.

(B) Storage and read-out are separate design choices. The Spatial read-out ablation shows this most directly: withholding the stored tokens makes Replay PSNR go up, not down. Raw context makes most of its open-domain gain from capacity under a trivial read-out, while block-wise State-Space gains from read-out structure at fixed capacity. The two families move along different axes of the storage-read-out plane, so a useful minimum is to report one ablation that fixes storage and varies read-out, and one that does the reverse.

(C) Compactness is not free. The two most aggressively compact mechanisms, Spatial Memory and hybrid compression, post the two lowest open-domain VLM scores (6.00 and 8.75), while raw context at $K=20$ reaches 58.63 and block-wise State-Space reaches 69.00. What separates the strong rows is not whether the memory is compact, but whether the compact representation is read out in a way that lets the generator put the same object back after the camera leaves it. Future compact designs should therefore be benchmarked against Context $K=20$ and block-wise State-Space on open-domain VLM, not against I2V.

These claims sit within clear limits. With eight cases per row, small open-domain VLM differences (within roughly five points) should not be over-interpreted, and the matrix uses a single training budget and a single dataset, so conclusions about which mechanism wins may shift at larger compute even if those about which axes matter are more durable. [section 7](#) returns to these points.

7 Discussion

We have presented Echo-Memory, a controlled study of memory in action world models. By fixing the backbone, optimizer, sampler, and evaluation protocol, the study makes the memory/context profile the main experimental variable. The resulting matrix separates a model’s ability to follow camera actions from its ability to preserve the world that those actions leave behind.

Main findings. Three lessons stand out. First, raw context is a strong baseline: increasing context improves open-domain revisit more reliably than it improves replay metrics, suggesting that capacity matters more than pixel fidelity for returning to the same object or scene. Second, compactness alone is not enough. Spatial Memory and hybrid compression are efficient, but without an object-aware read-out they lose the identity evidence needed by the generator. Third, implicit memory depends on structure. The State-Space variants differ most when recurrence is made non-bypassable, not merely lightweight.

Evaluation lesson. Replay quality and revisit quality are not monotonically aligned. Several rows that look competitive under low-level replay metrics fall behind under semantic revisit scores. Selecting models by Replay alone would therefore favor the wrong family for interactive, revisitable world generation. This is why we report replay, in-domain return, and open-domain return as separate evidence rather than collapsing them into one scalar.

Limitations. The study is still bounded in scope. We train on a single dataset, so the ranking of mechanisms may change with noisier poses, different camera statistics, larger compute, or multi-stage curricula. The open-domain scores also rely on a VLM judge; our cross-judge check (section 5.5) is encouraging, but broader human calibration would make the comparison stronger. Finally, revisit quality is not yet available as a cheap training-time signal, so model selection still has to lean on imperfect proxies.

Outlook. The next step is to make memory design explicitly revisit-aware: compression should preserve high-value object evidence, spatial summaries should expose object-level read-out to the generator, and structured recurrence should be evaluated as a first-class design axis. More broadly, Echo-Memory argues that world models should be measured not only by how smoothly they continue a trajectory, but by whether they can return to the same world after leaving it.

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